# C Programming Exam Questions - Complete Guide

## 1. Structure Program - Employee Salary Management

**Question:** Write a program using structure to store the name, id, age and salary of 3 employees and display the details of the employee with the highest salary.

```
#include &lt;stdio.h&gt;

struct employee {
    char name[25];
    int id;
    int age;
    int salary;
};

int main() {
    struct employee emp[3];
    printf("Enter the details of each employee\n");

    for(int i = 0; i &lt; 3; i++) {
        printf("Enter the name of employee: ");
        scanf("%s", emp[i].name);
        printf("Enter the id of employee: ");
        scanf("%d", &amp;emp[i].id);
        printf("Enter the age of employee: ");
        scanf("%d", &amp;emp[i].age);
        printf("Enter the salary of employee: ");
        scanf("%d", &amp;emp[i].salary);
        printf("\n");
    }

    // Find employee with highest salary
    int maxIndex = 0;
    for(int i = 1; i &lt; 3; i++) {
        if(emp[i].salary &gt; emp[maxIndex].salary) {
            maxIndex = i;
        }
    }

    // Display the employee with highest salary
    printf("Employee with the highest salary:\n");
    printf("Name: %s\n", emp[maxIndex].name);
    printf("ID: %d\n", emp[maxIndex].id);
    printf("Age: %d\n", emp[maxIndex].age);
    printf("Salary: %d\n", emp[maxIndex].salary);

    return 0;
}
```

## 2. Factorial Using Recursion

**Question:** WAP to calculate factorial of a number using recursion.

```
#include &lt;stdio.h&gt;

int factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    } else {
        return n * factorial(n - 1);
    }
}

int main() {
    int num, result;
    printf("Enter a number: ");
    scanf("%d", &amp;num);

    if (num &lt; 0) {
        printf("Factorial is not defined for negative numbers.\n");
    } else {
        result = factorial(num);
        printf("Factorial of %d is %d\n", num, result);
    }

    return 0;
}
```

## 3. Fibonacci Series Using Recursion

**Question:** Write a program to display the fibonacci series using recursive function.

```
#include &lt;stdio.h&gt;

int fibo(int n) {
    if (n &lt; 1) {
        return 0;
    } else if(n == 1) {
        return 1;
    } else {
        return fibo(n-1) + fibo(n-2);
    }
}

int main() {
    int n;
    printf("Enter the number of terms: ");
    scanf("%d", &amp;n);

    printf("Fibonacci series: ");
    for(int i = 0; i &lt; n; i++) {
        printf("%d ", fibo(i));
```

```
    }
    printf("\n");

    return 0;
}
```

### 4. Student Structure with Pass/Fail Analysis

**Question:** Write a program using structure to store student details and display passed students.

```
#include &lt;stdio.h&gt;

struct student {
    char name[20];
    int roll;
    int marks;
};

int main() {
    struct student std[3];
    printf("Enter Students Details\n");

    for(int i = 0; i &lt; 3; i++) {
        printf("Enter the name of student %d: ", i+1);
        scanf("%s", std[i].name);
        printf("Enter the Roll of %s: ", std[i].name);
        scanf("%d", &amp;std[i].roll);
        printf("%s's marks: ", std[i].name);
        scanf("%d", &amp;std[i].marks);
        printf("\n");
    }

    int totalPassed = 0;
    printf("Passed students\n");
    printf("Name\tRoll\tMarks\n");

    for (int i = 0; i &lt; 3; i++) {
        if (std[i].marks &gt;= 40) {
            printf("%s\t%d\t%d\n", std[i].name, std[i].roll, std[i].marks);
            totalPassed++;
        }
    }

    printf("Total Passed Students = %d\n", totalPassed);
    return 0;
}
```

## 5. Circle Area and Circumference Calculator

**Question:** Write a program using user defined functions to calculate circumference and area of circle.

```
#include &lt;stdio.h&gt;
#define PI 3.14159

float calculateCircumference(float radius) {
    return 2 * PI * radius;
}

float calculateArea(float radius) {
    return PI * radius * radius;
}

int main() {
    float radius;
    printf("Enter the radius of circle: ");
    scanf("%f", &amp;radius);

    printf("Circumference: %.2f\n", calculateCircumference(radius));
    printf("Area: %.2f\n", calculateArea(radius));

    return 0;
}
```

## 6. Positive/Negative Number Check

**Question:** Write a program to check whether a number is positive or negative using function.

```
#include &lt;stdio.h&gt;

void checkPositive(int n) {
    if (n &gt;= 0) {
        printf("The entered number is positive\n");
    } else {
        printf("The entered number is negative\n");
    }
}

int main() {
    int n;
    printf("Enter the number: ");
    scanf("%d", &amp;n);
    checkPositive(n);
    return 0;
}
```

## 7. Odd or Even Check Using Function

**Question:** Write a program to check the number is odd or even using function.

```
#include &lt;stdio.h&gt;

void checkOddEven(int n) {
    if (n % 2 == 0) {
        printf("%d is even\n", n);
    } else {
        printf("%d is odd\n", n);
    }
}

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &amp;n);
    checkOddEven(n);
    return 0;
}
```

## 8. Prime Number Check Using Function

**Question:** Write a program to check if the number is prime or not using function.

```
#include &lt;stdio.h&gt;
#include &lt;math.h&gt;

int isPrime(int n) {
    if (n &lt;= 1) return 0;
    for (int i = 2; i &lt;= sqrt(n); i++) {
        if (n % i == 0) return 0;
    }
    return 1;
}

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &amp;n);

    if (isPrime(n)) {
        printf("%d is a prime number\n", n);
    } else {
        printf("%d is not a prime number\n", n);
    }

    return 0;
}
```

## 9. Palindrome Check

**Question:** Write a program to check palindrome.

```
#include &lt;stdio.h&gt;
#include &lt;string.h&gt;

int main() {
    char str[100];
    int length, i, flag = 1;

    printf("Enter a string: ");
    scanf("%s", str);

    length = strlen(str);

    for(i = 0; i &lt; length / 2; i++) {
        if(str[i] != str[length - i - 1]) {
            flag = 0;
            break;
        }
    }

    if(flag)
        printf("%s is a palindrome\n", str);
    else
        printf("%s is not a palindrome\n", str);

    return 0;
}
```

## 10. Array Sorting (Ascending and Descending)

**Question:** Write a program to sort an array in ascending and descending order.

```
#include &lt;stdio.h&gt;

void bubbleSort(int arr[], int n, int ascending) {
    for (int i = 0; i &lt; n-1; i++) {
        for (int j = 0; j &lt; n-i-1; j++) {
            if ((ascending &amp;&amp; arr[j] &gt; arr[j+1]) ||
                (!ascending &amp;&amp; arr[j] &lt; arr[j+1])) {
                int temp = arr[j];
                arr[j] = arr[j+1];
                arr[j+1] = temp;
            }
        }
    }
}

void printArray(int arr[], int n) {
    for (int i = 0; i &lt; n; i++) {
        printf("%d ", arr[i]);
```

```
    }
    printf("\n");
}

int main() {
    int n;
    printf("Enter number of elements: ");
    scanf("%d", &amp;n);

    int arr[n];
    printf("Enter elements: ");
    for (int i = 0; i &lt; n; i++) {
        scanf("%d", &amp;arr[i]);
    }

    printf("Original array: ");
    printArray(arr, n);

    bubbleSort(arr, n, 1);
    printf("Ascending order: ");
    printArray(arr, n);

    bubbleSort(arr, n, 0);
    printf("Descending order: ");
    printArray(arr, n);

    return 0;
}
```

## 11. Power of Number Using Function

**Question:** WAP to find power of a number using function.

```
#include &lt;stdio.h&gt;

int power(int base, int exp) {
    if (exp == 0) return 1;
    return base * power(base, exp - 1);
}

int main() {
    int base, exp;
    printf("Enter base: ");
    scanf("%d", &amp;base);
    printf("Enter exponent: ");
    scanf("%d", &amp;exp);

    printf("%d^%d = %d\n", base, exp, power(base, exp));
    return 0;
}
```

## 12. Cricket Strike Rate Calculator

**Question:** Write a program to enter name of Batters, Runs scored and Balls faced using structure and display their strike rate.

```
#include &lt;stdio.h&gt;

struct batter {
    char name[50];
    int runs;
    int balls;
    float strikeRate;
};

int main() {
    int n;
    printf("Enter number of batters: ");
    scanf("%d", &amp;n);

    struct batter batters[n];

    for (int i = 0; i &lt; n; i++) {
        printf("Enter name of batter %d: ", i+1);
        scanf("%s", batters[i].name);
        printf("Enter runs scored: ");
        scanf("%d", &amp;batters[i].runs);
        printf("Enter balls faced: ");
        scanf("%d", &amp;batters[i].balls);

        batters[i].strikeRate = (float)batters[i].runs / batters[i].balls * 100;
    }

    printf("\nBatter Details:\n");
    printf("Name\t\tRuns\tBalls\tStrike Rate\n");
    for (int i = 0; i &lt; n; i++) {
        printf("%s\t\t%d\t%d\t%.2f\n", batters[i].name,
               batters[i].runs, batters[i].balls, batters[i].strikeRate);
    }

    return 0;
}
```

## 13. Swap Two Numbers Using Pointers

**Question:** Write a program to swap two numbers using pointers.

```
#include &lt;stdio.h&gt;

void swap(int *a, int *b) {
    int temp = *a;
    *a = *b;
    *b = temp;
}
```

```
int main() {
    int x, y;
    printf("Enter two numbers: ");
    scanf("%d %d", &amp;x, &amp;y);

    printf("Before swapping: x = %d, y = %d\n", x, y);
    swap(&amp;x, &amp;y);
    printf("After swapping: x = %d, y = %d\n", x, y);

    return 0;
}
```

## 14. Greatest Number in Array Using Functions

**Question:** Write a program to take an array of integers and display the greatest number using functions.

```
#include &lt;stdio.h&gt;

int findGreatest(int arr[], int n) {
    int max = arr[0];
    for (int i = 1; i &lt; n; i++) {
        if (arr[i] &gt; max) {
            max = arr[i];
        }
    }
    return max;
}

int main() {
    int n;
    printf("Enter number of elements: ");
    scanf("%d", &amp;n);

    int arr[n];
    printf("Enter elements: ");
    for (int i = 0; i &lt; n; i++) {
        scanf("%d", &amp;arr[i]);
    }

    printf("Greatest number: %d\n", findGreatest(arr, n));
    return 0;
}
```

## 15. Multiplication Table

**Question:** Write a program to display multiplication table of "n".

```
#include &lt;stdio.h&gt;

void multiplicationTable(int n) {
    printf("Multiplication table of %d:\n", n);
    for (int i = 1; i &lt;= 10; i++) {
        printf("%d x %d = %d\n", n, i, n * i);
    }
}

int main() {
    int n;
    printf("Enter a number: ");
    scanf("%d", &amp;n);
    multiplicationTable(n);
    return 0;
}
```

## 16. Quadratic Equation Roots

**Question:** WAP to take inputs of a, b, c for equation $ax^2 + bx + c$ and display the roots.

```
#include &lt;stdio.h&gt;
#include &lt;math.h&gt;

int main() {
    float a, b, c, discriminant, root1, root2;

    printf("Enter coefficients a, b, c: ");
    scanf("%f %f %f", &amp;a, &amp;b, &amp;c);

    discriminant = b * b - 4 * a * c;

    if (discriminant &gt; 0) {
        root1 = (-b + sqrt(discriminant)) / (2 * a);
        root2 = (-b - sqrt(discriminant)) / (2 * a);
        printf("Roots are real and different.\n");
        printf("Root1 = %.2f, Root2 = %.2f\n", root1, root2);
    } else if (discriminant == 0) {
        root1 = -b / (2 * a);
        printf("Roots are real and same.\n");
        printf("Root1 = Root2 = %.2f\n", root1);
    } else {
        printf("Roots are complex and different.\n");
        printf("Root1 = %.2f + %.2fi\n", -b/(2*a), sqrt(-discriminant)/(2*a));
        printf("Root2 = %.2f - %.2fi\n", -b/(2*a), sqrt(-discriminant)/(2*a));
    }
```

```
    return 0;
}
```

### Practice Tips for Exam:

1. **Understand the logic** behind each program
2. **Practice writing code** without looking at examples
3. **Focus on syntax** - proper headers, data types, brackets
4. **Test your programs** mentally by tracing through execution
5. **Remember common functions** like strlen(), sqrt(), etc.
6. **Practice structure and array** programs as they are frequently asked
7. **Understand pointer concepts** for swapping programs